



Tymtix team (left to right): Jyoti Chauhan, Vinayak Hogde, Mahesh T., Kalaiselvan Vallal, Banuja Nayak, Yuvaraj Tana, Prathima D., Tejas J., Veena Putham, Vipin E., Aparna Rajan, Prabu Surendra, Ragavendiran E., Kailash Dahiya and Krishnaveni

from a large number of SensAiry-enabled vehicles to judge the road conditions of different cities. This idea has the potential to change the way we think about judging road quality in cities.

How it stacks up to the alternatives

There are a lot of alternative tyre pressure-monitoring systems (TPMSes) in the market where they connect a gadget on the tyre and a small display is mounted on the dashboard through which you can see data on tyre pressure. In high-end vehicles this could be integrated to the vehicle's stock infotainment system, too. But these solutions require vehicle owners to be inside their vehicles to check tyre pressure, which is not a requirement for SensAiry.

One unique challenge that none of the existing solutions for non-luxury vehicles solve is that these do not alert you about a tyre puncture or critically-low tyre pressure until you are near the car or turn on the ignition. The problem here is that it makes the situation ripe for that unfortunate day where you are greeted by a flat tyre while hurrying to your office in a jacket and trousers.

A more significant benefit is for fleet-management companies that have long trailers or road trains with a large number of tyres at complex

locations. Some might have multi-axle tyres where there are eight tyres at one point in a bus. Then there are lorries with 64 tyres—imagine the pain of installing a normal TPMS.

Design challenges

Tyres on any vehicle face some very extreme conditions, so placing your components inside there is a technically-challenging aspect of their design. In fact, some people say that this is exactly the place where an electronic system should not be installed.

Breaking out of enclosure problems. Surendra explains, "One of our earliest problems was that we were unable to fit the sensing system to properly measure pressure in a feasible way. We could get it in, through a laborious process, but it would not make sense to do it from a customer standpoint or even allow us to do a proper demonstration. So while we were trying to figure this out, one of our sensor vendor's chief research and development person invited me over and showed the facilities that they have. It was during this tour that I discovered the kind of test setup we actually needed to build the product we had in mind."

Another problem that they had was that the enclosure was built out of aluminium. Now, aluminium has this problem that it absorbs all sig-

nals due to it acting like a Faraday's cage. When they placed the sensor inside it, no signals could make their way out of the device. "The team was very disappointed when this happened," says Surendra. "While we were trying to figure out how to solve this problem, we came upon a material that had the characteristics of ABS plastic and did not absorb the signals that our device needed to transmit."

The omnipresent battery life problem. Battery life is another area where they did some tuning and revisions to come up with the perfect sauce. For instance, they designed a range extender in the device that has the consequence of reducing battery life. So what they did was modify the device so that it only activates the extender when it is really needed, like in the case of a puncture where the car needs to alert the user as early as possible.

Another event that deactivates the range extender is when the car is moving. So they used the accelerometer to figure out if the car was moving and deactivate it if it was. It also changes the frequency of testing depending on different times of the day or usage history. They eventually left the range extender out of the first version of the product to control cost, but there is a good chance that it would show up in a future version.

"Testing is one area where I am facing a challenge. Normal tyres can only handle up to 50psi of pressure, but I needed to test the sensor at 150psi to make sure it works in heavy-duty tyres. I also needed to test the sensors while running at 10,000rpm. So what I did was to get access to a dynamometer; it was pretty expensive in our case since we need to constantly try and improve the system. We are now trying to set up a custom jig to get this done. We might go for a gyroscope instead of an accelerometer for suspension and wheel-alignment issues in a future product," explains Surendra. ●